

coils do not burn out. Even with the higher price tag, however, people prefer purchasing passive speakers so they can have more fine-tuned control over their audio systems.

In contrast, active, or powered speakers, contain active components in their crossover filter. The most commonly used active component is the op-amp. Unlike passive crossovers, which operate after the power amplifier's output, at high current and in some cases high voltage, active crossovers are operated at levels suited to power amplifier inputs. In addition, active speakers contain an internal amplifier for the speaker. However, multi-channel speakers, that contain multiple drivers, require one amplifier per driver. This might make active speaker systems slightly bulkier and expensive. However, their inherent advantage lies in the fact that this separation of amplifiers allows for better isolation of each driver from signals handled by other drivers, thus reducing noise, distortion and overdriving in the audio produced. Moreover, the absence of loss of energy from passive components means that amplifier requirements are reduced considerably, thus reducing the overall power the speaker system would use.

Overall, when it comes to choosing between active and passive speakers, the decision should be made seeing the final usage requirements. In the case you're looking for a home speaker system, you will notice a higher cost for active systems because of the additional amplification channels required. However, if you are looking for extremely high fidelity, it is worth considering active speakers over passive ones. You'll also end up eliminating the hassle of having to choose an external amplifier for your passive system, unless you're actually looking to be able to customise the amp being used all by yourself. Though none of our readers will probably be looking for speakers to use in concerts, a word of advice: passive systems, which always end up being cheaper and more compact than active systems are your best bet, hands down. Just saying.

Channel configuration:

This is a particular area of interest for audiophiles looking to set up their loudspeaker systems at home, mostly for the purpose of a home theatre system.

Let's first get the definition of channels out of the way. When we refer to an audio channel, we enter the domain of sound mixing and the storage of analog audio signals in a digital format. While the channel refers to the path taken by the signal, in clear and precise language, the context of audio gives it another meaning. It is best understood by looking at the final product and